

Time Series Project: IPI Food Industries

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1 Part I: The data

1.1 1. What does the chosen series represent?

This dataset is an Industrial Production Index (IPI) for the food industries sector in France, monthly frequency, seasonally and working-day adjusted (CVS-CJO), base 100 in 2021. The variable is an index, so it is unit-free. Because the series is already CVS-CJO, I do not apply additional seasonal adjustment.

```
meta_candidates <- list.files("data", pattern = "caract", full.names = TRUE)
if (length(meta_candidates) == 0) {
  stop("Metadata file not found in data/.")
}
meta_path <- meta_candidates[1]
meta <- read_delim(
  meta_path,
  delim = ";",
  col_names = TRUE,
  show_col_types = FALSE
)
names_norm <- iconv(names(meta), from = "", to = "ASCII//TRANSLIT")
pick_col <- function(pattern) {
```

```

idx <- which(grepl(pattern, names_norm, ignore.case = TRUE))
if (length(idx) == 0) return(NA_character_)
names(meta)[idx[1]]
}

cols <- c(
  pick_col("idbank"),
  pick_col("mise"),
  pick_col("period"),
  pick_col("unit"),
  pick_col("activ"),
  pick_col("correct"),
  pick_col("indicat"),
  pick_col("zone"),
  pick_col("base")
)
cols <- cols[!is.na(cols)]

meta %>%
  slice(1) %>%
  select(all_of(cols)) %>%
  kable()

```

idBank	Dernière mise à jour	Périodicité	Unité	Activité	Correction	Indicateur	Zone géographique	Base de l'indice
01076763	2020/04/20 08:45	Mensuelle	heures	10 - Industries alimentaires	Corrigé des variations saisonnières et du nombre de jours ouvrables	Indice de la production industrielle	France métropolitaine	2021

Note: if column names in the metadata file contain accents on your machine, the table above may require manual renaming, but the content is the same.

1.2 2. Transform the series to make it stationary if necessary

We read the monthly values file, which contains multiple vintages. Each vintage is stored as a pair of columns: a value column followed by a code column (A, P, R, etc.). The row labeled “Mises a jour” gives the update date for each vintage. We select the most recent vintage by default.

```

values_path <- file.path("data", "valeurs_mensuelles.csv")
raw <- read_delim(
  values_path,
  delim = ";",
  col_names = FALSE,
  show_col_types = FALSE
)

row_label <- raw[[1]]
row_label_norm <- iconv(row_label, from = "", to = "ASCII//TRANSLIT")
row_label_norm <- trimws(row_label_norm)

```

```

idx_updates <- which(row_label_norm == "Mises a jour")[1]
idx_period <- which(row_label_norm == "Periode")[1]
if (is.na(idx_updates) || is.na(idx_period)) {
  stop("Could not locate 'Mises a jour' or 'Periode' rows in the values file.")
}

updates_row <- raw[idx_updates, , drop = FALSE]
update_strings <- as.character(unlist(updates_row[-1]))
update_dates <- suppressWarnings(lubridate::dmy_hm(update_strings, quiet = TRUE))

value_cols <- which(!is.na(update_dates)) + 1
vintage_tbl <- tibble(
  vintage_id = seq_along(value_cols),
  update_date = update_dates[!is.na(update_dates)],
  value_col = value_cols
) %>% arrange(update_date)

vintage_tbl %>% tail(6) %>% kable()

```

vintage_id	update_date	value_col
6	2025-11-05 08:45:00	12
5	2025-12-05 08:45:00	10
4	2026-01-09 08:45:00	8
3	2026-02-05 08:45:00	6
2	2026-03-05 08:45:00	4
1	2026-04-03 08:45:00	2

```

latest_vintage <- vintage_tbl %>% slice_max(update_date, n = 1)
chosen_col <- latest_vintage$value_col

series_raw <- raw[(idx_period + 1):nrow(raw), , drop = FALSE] %>%
  transmute(
    period = .data[[names(raw)[1]]],
    value = .data[[names(raw)[chosen_col]]]
  ) %>%
  mutate(
    date = as.Date(paste0(period, "-01")),
    value = readr::parse_number(value, locale = locale(decimal_mark = "."))
  ) %>%
  filter(!is.na(date))

start_year <- year(min(series_raw$date))
start_month <- month(min(series_raw$date))

Y <- ts(series_raw$value, start = c(start_year, start_month), frequency = 12)
Y <- stats::na.omit(Y)

summary(series_raw$value)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  83.30   95.65   97.98   97.14   99.43  103.24

```

We consider a log transform if all values are strictly positive. Then we determine the number of differences needed using unit-root tests and verify stationarity with ADF and KPSS.

```
use_log <- all(Y > 0, na.rm = TRUE)
Z <- if (use_log) log(Y) else Y

# Suggested differencing from ADF and KPSS
nd_adf <- forecast::ndiffs(Z, alpha = 0.05, test = "adf")
nd_kpss <- forecast::ndiffs(Z, alpha = 0.05, test = "kpss")
d <- max(nd_adf, nd_kpss)

X <- diff(Z, differences = d)

adf_Z <- tseries::adf.test(Z)
kpss_Z <- tseries::kpss.test(Z, null = "Level")

adf_X <- tseries::adf.test(X)
kpss_X <- tseries::kpss.test(X, null = "Level")

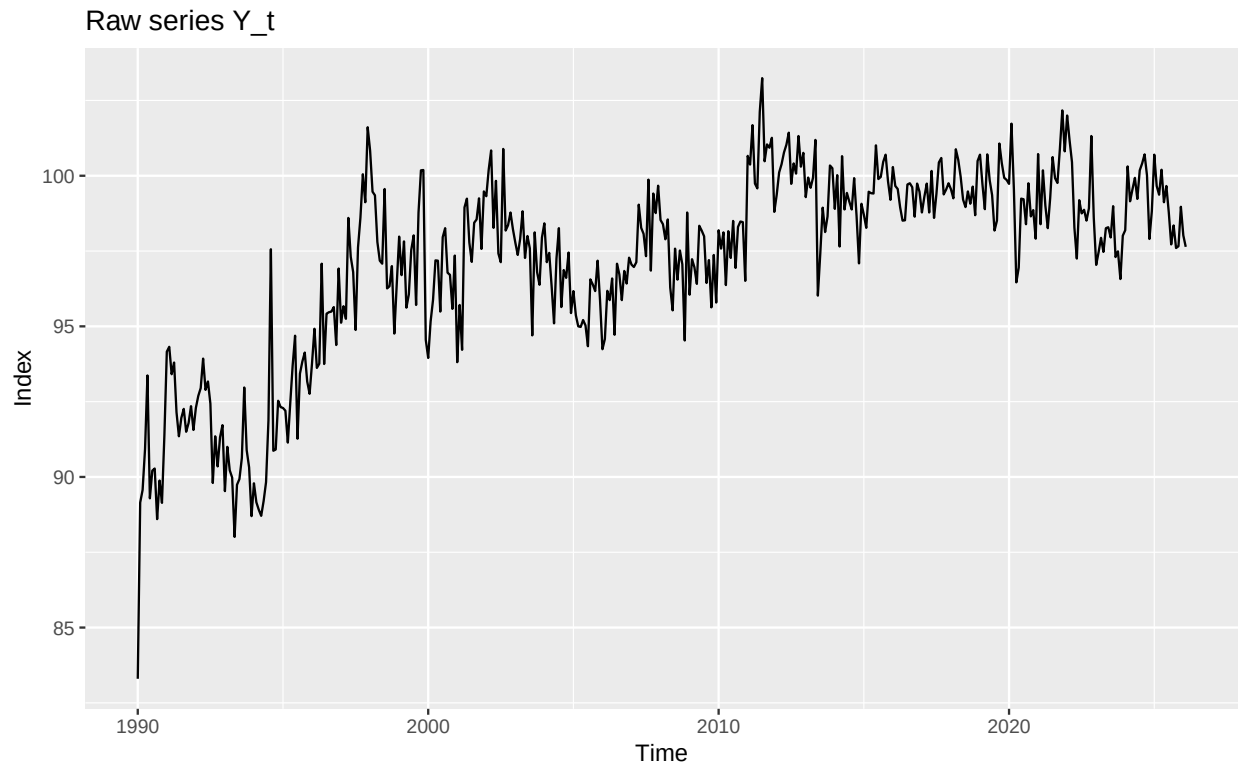
list(
  use_log = use_log,
  d_adf = nd_adf,
  d_kpss = nd_kpss,
  d_chosen = d,
  adf_Z_p = adf_Z$p.value,
  kpss_Z_p = kpss_Z$p.value,
  adf_X_p = adf_X$p.value,
  kpss_X_p = kpss_X$p.value
)
```

```
## $use_log
## [1] TRUE
##
## $d_adf
## [1] 0
##
## $d_kpss
## [1] 1
##
## $d_chosen
## [1] 1
##
## $adf_Z_p
## [1] 0.3093639
##
## $kpss_Z_p
## [1] 0.01
##
## $adf_X_p
## [1] 0.01
##
## $kpss_X_p
## [1] 0.1
```

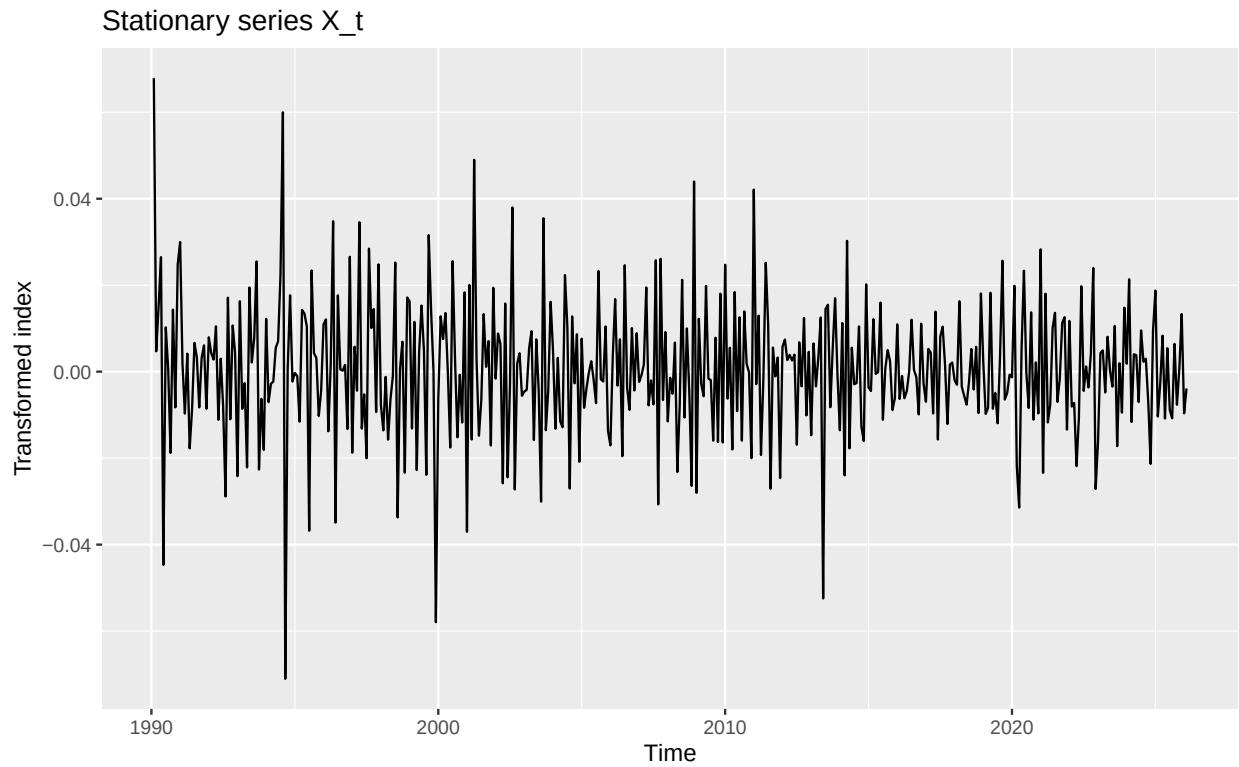
We define the stationary series as $X_t = \nabla^d Z_t$ where $Z_t = \log Y_t$ if all values are positive, otherwise $Z_t = Y_t$. This choice is justified by the unit-root tests above (ADF should reject non-stationarity and KPSS should not reject stationarity on X_t).

1.3 3. Graphical representation before and after transformation

```
autoplot(Y) +  
  labs(title = "Raw series Y_t", x = "Time", y = "Index")
```



```
autoplot(X) +  
  labs(title = "Stationary series X_t", x = "Time", y = "Transformed index")
```

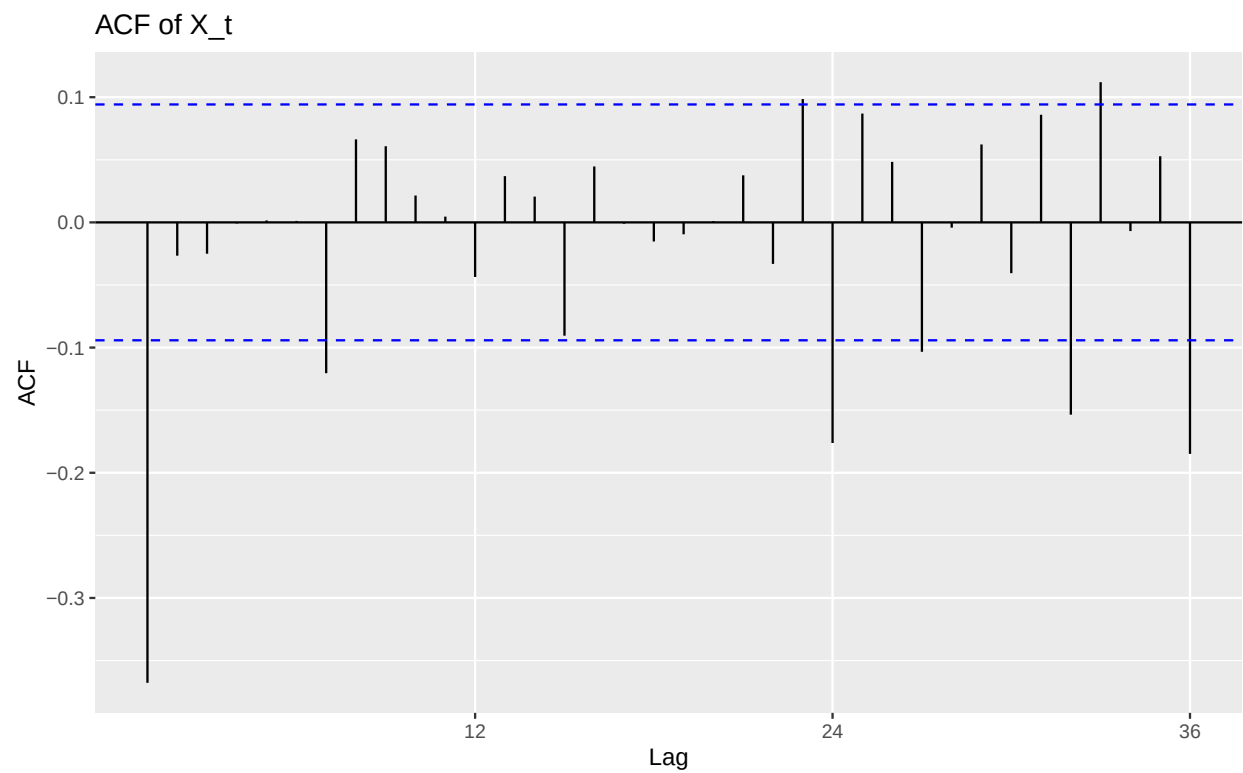


2 Part II: ARMA models

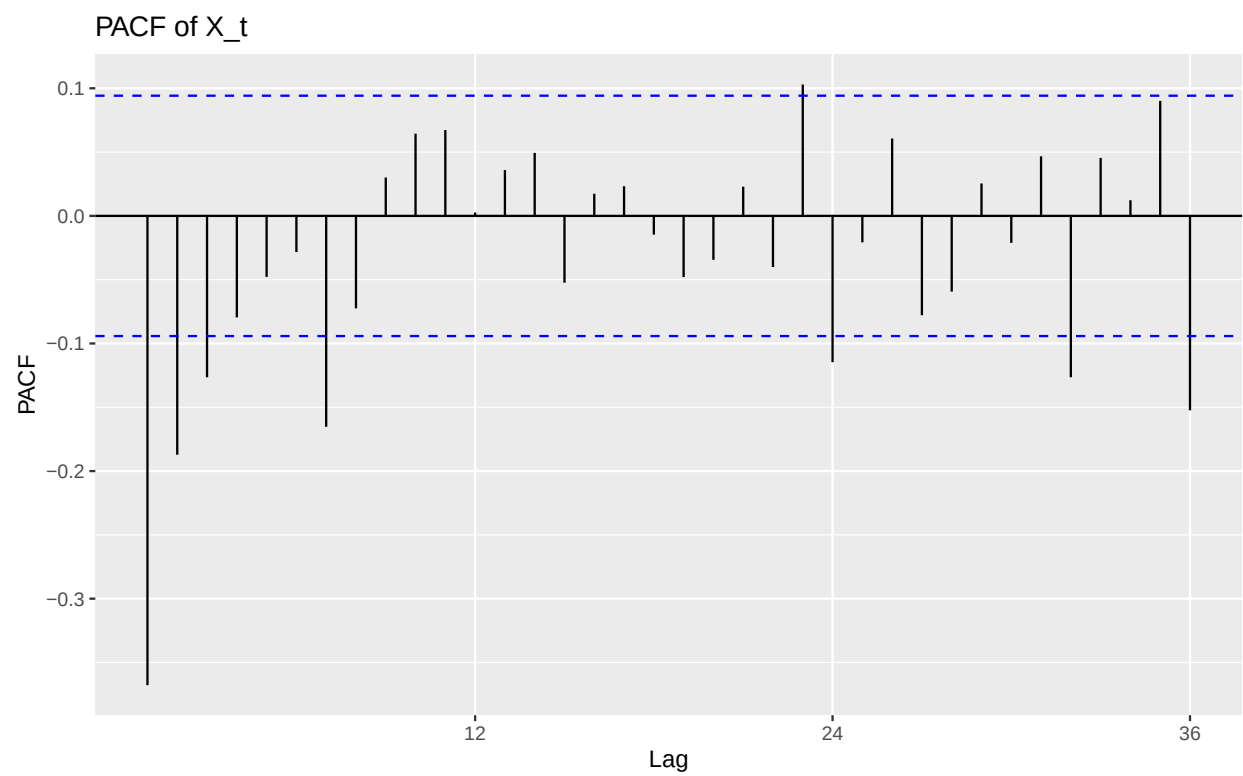
2.1 4. Pick and estimate an ARMA(p,q) model for X_t

We use ACF/PACF diagnostics and an information-criterion search on a small grid to select a parsimonious ARMA model.

```
ggAcf(X, lag.max = 36) + labs(title = "ACF of  $X_t$ ")
```



```
ggPacf(X, lag.max = 36) + labs(title = "PACF of  $X_t$ ")
```



```

max_p <- 5
max_q <- 5

grid <- expand.grid(p = 0:max_p, q = 0:max_q) %>%
  rowwise() %>%
  mutate(
    fit = list(tryCatch(
      Arima(X, order = c(p, 0, q), include.mean = TRUE),
      error = function(e) NULL
    )),
    aic = ifelse(is.null(fit), NA, AIC(fit)),
    bic = ifelse(is.null(fit), NA, BIC(fit))
  ) %>%
  ungroup() %>%
  arrange(aic)

grid %>%
  filter(!is.na(aic)) %>%
  head(8) %>%
  kable()

```


p	q	fit	aicbic
p	q	fit	aicbic
4	5	0.05410488, 0.6039334, 0.01323962, -0.9644772, -0.6250231, -0.6444369, 0.3517274, 0.9961359, -0.6082907, 0.0002556828, 0.0001839512, 0.0002584824, -0.0001298926, -0.0001537762, 8.031683e-05, -0.0002867987, 0.0001549265, 0.0003663061, -2.338742e-05, -0.0003388179, 2.657224e-08, -0.0001298926, 0.0003839223, 0.0001486754, 2.384752e-05, 0.0003063268, -0.0004707981, -0.0004876004, 0.0001304974, 0.000552504, -1.918943e-08, -0.0001537762, 0.0001486754, 0.0007142125, -2.843381e-05, 0.0006050051, -0.0006269226, -0.00122967, 0.0001628178, 0.001185888, -6.88399e-08, 8.031683e-05, 2.384752e-05, -2.843381e-05, 0.0002454927, 0.0001493414, -0.0002724473, -0.0002307518, -6.535543e-06, 0.0002822839, -2.646425e-08, -0.0002867987, 0.0003063268, 0.0006050051, 0.0001493414, 0.003245557, -0.00170602, -0.003172077, 0.001263277, 0.002738805, 7.902599e-08, 0.0001549265, -0.0004707981, -0.0006269226, -0.0002724473, -0.00170602, 0.002089751, 0.002290003, -0.00147994, -0.001486367, 1.611554e-07, 0.0003663061, -0.0004876004, -0.00122967, -0.0002307518, -0.003172077, 0.002290003, 0.004091193, -0.001239194, -0.003633718, 1.057553e-07, -2.338742e-05, 0.0001304974, 0.0001628178, -6.535543e-06, 0.001263277, -0.00147994, -0.001239194, 0.002336817, -0.0005338482, -4.591321e-08, -0.0003388179, 0.000552504, 0.001185888, 0.0002822839, 0.002738805, -0.001486367, -0.003633718, -0.0005338482, 0.005013246, -5.885178e-08, 2.657224e-08, -1.918943e-08, -6.88399e-08, -2.646425e-08, 7.902599e-08, 1.611554e-07, 1.057553e-07, -4.591321e-08, -5.885178e-08, 5.768054e-08, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, 1249.692, -2477.383, 4, 5, 0, 0, 12, 0, 0, 0.05654375, 0.02746041, 0.02900446, 0.03750521, -0.01942546, 0.006415773, 0.006918671, -0.004869014, 0.01073166, -0.003995085, 0.01958776, 0.028756, 0.01704967, -0.003017244, 0.00508638, -0.004489509, -0.006851495, 0.01174825, 0.009599134, -0.006192407, -0.004750152, -0.008681047, -0.01636246, -0.006160854, 0.001134364, 0.01127916, 0.01944977, 0.009262789, 0.01060809, -0.003521827, -0.03067095, -0.01047438, -0.0214315, -0.006412875, -0.004227486, -0.02146906, 0.004329202, -0.001015884, 0.002205202, -0.02263085, 0.007866924, -0.0008318153, 0.003053306, 0.02361406, -0.01398734, -0.008129824, -0.0230712, 0.005236677, 0.0005095384, -0.004396211, -0.002414965, -0.004603262, 0.0007425256, 0.01864973, 0.06501124, -0.02642097, -0.01116352, 0.01574224, 0.01470487, 0.00871178, 0.004603349, -0.01339426, -0.0007313503, 0.008539802, 0.01049289, -0.02852121, 0.01024298, 0.01329334, 0.01912059, 0.002211784, -0.001987943, 0.007419106, 0.008868263, -0.01064043, -0.01021921, 0.02786107, -0.01519898, 0.01237408, 0.01296178, 0.01124137, 0.00969749, -0.009346751, 0.01727253, -0.01390048, -0.005075703, -0.01009178, 0.02933451, 0.007279122, 0.003476939, -0.01357628, 0.02174224, 0.02203382, 0.02527424, 0.001567099, 0.02301593, 0.003799288, -0.009305097, -0.003531005, -0.01494104, -0.01055565, -0.008102221, 0.01903977, -0.02636672, -0.01935824, -0.008807222, -0.03168274, 1.750857e-05, 0.01568106, 0.0006474571, 0.0155458, -0.01384827, -0.0001627695, 0.009847907, 0.008561498, -0.02070731, 0.01424575, 0.02392933, 0.01422134, -0.04512957, -0.02800565, -0.005371375, 0.006546134, 0.01313726, 0.002742401, -0.01758315, 0.009400316, 0.009730996, -0.007158157, -0.002032708, -0.007532445, 0.01474018, -0.02752266, 0.002403638, -0.0213898, 0.03414335, 0.01657591, -0.004288907, -0.005441382, 0.01062771, 0.01609656, 0.01761529, -0.004499018, 0.01728483, 0.0006218626, 0.009310492, 0.004997736, -0.02386716, 0.00582823, -0.023684, -0.007023926, 0.0321955, -0.008028692, 0.001449306, -0.006254541, -0.006435452, -0.01459875, -0.0165652, 0.001778103, 0.004249096, -0.003984547, 0.005279986, -0.002769973, -0.02541072, 0.01063739, -0.006506148, -0.01124354, 0.003083398, 0.01174106, -0.01018462, 0.003380734, -0.006885501, -0.01790711, 0.01700613, 0.01199668, -0.0184577, -0.001597185, -0.01070678, 0.008292677, -0.02168906, 0.001340157, -0.005593074, -0.009310939, 0.001008196, -0.005861379, -0.003611989, -0.0132614, 0.008940273, 0.008241206, -0.002572173, 0.01528426, -0.0025167, -0.0181569, -0.0009289507, 0.008774022, 0.00358803, 0.005344365, -0.02254202, 0.01561378, -0.001481951, -0.002315982, 0.01077023, 0.0006516255, 0.01824582, 0.0003067864, 0.002966512, -8.187901e-05, 0.0121917, 0.004884612, -0.006314348, -0.005074304, 0.02475665, -0.01806751, 0.02536964, -0.0005887715, 0.01342756, -0.006371582, -0.01264294, -0.005747617, -0.006563605, -0.02069977, -0.01889979, 0.005461697, 0.001218423, 0.002664413, -0.0004448613, -0.0276857, 0.02085803, -0.01134741, 0.000765401, -1.24536e-05, -0.003340506, 0.01571879, 0.01234009, 0.002784761, -0.01353055, -6.570955e-05, -0.02128401, 0.008217814, -0.01661286, 0.01637254, 0.003813392, 0.005965939, -0.00913639, 0.01004285, -0.001135322, 0.01252516, -0.01276854, 0.009605137, 0.002053566, 0.002201605, -0.01743685, 0.02871232, 0.01825957, 0.0226057, -0.004037057, -0.001641659, 0.02120531, 0.02534006, -0.01384512, -0.002639719, -0.002777649, -0.0002306775, -0.02249312,	- - 2477.383, 5

p	q	fit	aicbic
5	5	0.2006429, -0.6665906, 0.1662615, -0.9225584, 0.2584308, -0.7433163, 0.7334929, -0.5567505, 0.9734253, -0.7774638, 0.0002527873, 0.0001829099, 0.005384125, 0.0004736066, 0.003827243, 0.000268768, 0.005056282, -0.002729373, 2.367107e-06, -0.002090123, -2.149795e-05, -0.002664149, -2.395382e-07, 0.0004736066, 0.0004672477, 0.0004084639, 9.31703e-05, 0.0003274019, -0.0003440307, -6.658982e-05, -0.0002012875, 1.960368e-05, -0.0003075468, -3.454056e-08, 0.003827243, 0.0004084639, 0.003090814, 0.0003418582, 0.003760351, -0.001998585, -5.565727e-05, -0.001508914, -6.296032e-05, -0.002009094, -1.675391e-07, 0.000268768, 9.31703e-05, 0.0003418582, 0.0004062553, 0.0003916257, -0.0002498171, 8.738473e-05, -0.0002630482, -1.150853e-05, -0.0002562182, -3.014847e-08, 0.005056282, 0.0003274019, 0.003760351, 0.0003916257, 0.005268582, -0.002532491, -6.897503e-05, -0.001939084, -0.0001343737, -0.002613507, -2.277077e-07, -0.002729373, -0.0003440307, -0.001998585, -0.0002498171, -0.002532491, 0.002583001, -0.0002073277, 0.00176625, -2.605074e-05, 0.00247231, 3.467214e-07, 2.367107e-06, -6.658982e-05, -5.565727e-05, 8.738473e-05, -6.897503e-05, -0.0002073277, 0.0004400666, -4.328395e-05, 0.0002276241, -0.0001926316, 4.824341e-09, -0.002090123, -0.0002012875, -0.001508914, -0.0002630482, -0.001939084, 0.00176625, -4.328395e-05, 0.00163415, -8.473854e-05, 0.001777913, 2.673454e-07, -2.149795e-05, 1.960368e-05, -6.296032e-05, -1.150853e-05, -0.0001343737, -2.605074e-05, 0.0002276241, -8.473854e-05, 0.0003973187, -0.000209534, 1.479347e-08, -0.002664149, -0.0003075468, -0.002009094, -0.0002562182, -0.002613507, 0.00247231, -0.0001926316, 0.001777913, -0.000209534, 0.002674831, 3.53391e-07, -2.395382e-07, -3.454056e-08, -1.675391e-07, -3.014847e-08, -2.277077e-07, 3.467214e-07, 4.824341e-09, 2.673454e-07, 1.479347e-08, 3.53391e-07, 4.526681e-08, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, 1249.597, -2475.193, 5, 5, 0, 0, 12, 0, 0, 0.05656881, 0.02835352, 0.03096272, 0.03934089, -0.01538492, 0.003493616, 0.002276312, -0.007917498, 0.00541455, -0.006332272, 0.02306142, 0.03259896, 0.02422227, 0.01092026, 0.008682927, -0.005559111, -0.01000067, -0.004769108, 0.005472184, -0.006768537, -0.004658418, 0.008390185, -0.007831872, -0.0001015873, 0.009871181, 0.004788771, 0.01122027, 0.001656225, -0.0005050888, -0.009016383, -0.026849, -0.003628588, -0.01310363, 0.006227698, -0.002013401, -0.02307653, 0.003979529, -0.0130954, -0.004256086, -0.02450634, 0.0001992836, 0.002464289, 0.005405116, 0.02378139, -0.002940729, -0.008402188, -0.02386936, 0.004218459, -0.01121699, -0.00963223, -0.004358313, -0.003836236, 0.003499147, 0.02707939, 0.06874324, -0.0287665, -0.00806573, 0.008608636, 0.006918976, 0.004466228, 0.001695757, -0.00789366, 0.003972574, 0.0133532, 0.02217696, -0.02473221, 0.01347593, 0.01231135, 0.004450024, -0.006623965, -0.003515304, 0.00498489, 0.01755432, -0.001524762, -0.004345403, 0.03360084, -0.01568126, 0.007047158, 0.01070954, 0.007556493, -0.000279939, -0.007156532, 0.01948684, -0.01260183, 0.008423634, -0.002478126, 0.03019741, 0.007567729, -0.004666425, -0.02270839, 0.02301493, 0.01743882, 0.02639731, 0.01008446, 0.02374182, 0.006177251, -0.002737054, -0.003405727, -0.01046189, -0.01026887, -0.01545884, 0.01670216, -0.02704886, -0.01704851, 0.003509722, -0.0230673, -0.002425963, 0.01740863, -0.009609124, 0.001724972, -0.01234115, -0.005333191, 0.008933394, 0.01676435, -0.02329051, 0.01723806, 0.02982408, 0.009884816, -0.04665084, -0.02224779, -0.01250402, 0.0006977056, 0.0172275, -0.0002665473, -0.01414838, 0.01850603, 0.004438793, -0.004515483, 0.0004800084, -0.01696499, 0.01265421, -0.03335301, -0.003297363, -0.0127467, 0.0397599, 0.01894629, 0.002897255, -0.009726334, 0.005964768, 0.01154612, 0.01243321, -0.006828339, 0.01953951, -0.0008463096, 0.008700151, 0.01622264, -0.01768348, 0.01207455, -0.01559398, -0.01933571, 0.02657605, -0.01168547, -0.008657039, 0.005907971, -0.001541338, 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p	q	fit	aicbic
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p	q	fit	aicbic
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p	q	fit	aicbic
3	4	-0.8567019, -0.6445592, 0.292901, 0.3340366, 0.1224924, -0.7701212, 0.02363142, 0.0002535027, 0.00018752, 0.0694461, 0.07962584, 0.06834286, -0.06924791, -0.03484482, -0.01984113, 0.04371547, -6.329453e-07, 0.07962584, 0.09144155, 0.07846152, -0.07939495, -0.04001756, -0.02279704, 0.05009739, -7.370535e-07, 0.06834286, 0.07846152, 0.06741312, -0.06812086, -0.03437706, -0.01956491, 0.04300949, -6.230752e-07, -0.06924791, -0.07939495, -0.06812086, 0.07152387, 0.03567397, 0.02007054, -0.04544194, 7.527385e-07, -0.03484482, -0.04001756, -0.03437706, 0.03567397, 0.01888111, 0.01113157, -0.02168412, 5.953725e-07, -0.01984113, -0.02279704, -0.01956491, 0.02007054, 0.01113157, 0.007020439, -0.01165224, 4.616193e-07, 0.04371547, 0.05009739, 0.04300949, -0.04544194, -0.02168412, -0.01165224, 0.02993115, -2.558487e-07, -6.329453e-07, -7.370535e-07, -6.230752e-07, 7.527385e-07, 5.953725e-07, 4.616193e-07, -2.558487e-07, 4.691393e-08, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, 1244.702, 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p	q	fit	aicbic
3	5	-1.845923, -1.786187, -0.6824124, 1.324697, 0.7444763, -0.4986156, -0.6504057, -0.155512, 0.0002551888, 0.0001874574, 0.235158, 0.270822, 0.2318404, -0.2327059, -0.1378015, -0.07108649, 0.1485249, 0.02089897, -1.024908e-06, 0.270822, 0.3119439, 0.2670304, -0.2680033, -0.1587454, -0.08189026, 0.1710188, 0.02408786, -1.188956e-06, 0.2318404, 0.2670304, 0.2286299, -0.2294091, -0.1358493, -0.07012885, 0.1463778, 0.0205581, -1.018562e-06, -0.2327059, -0.2680033, -0.2294091, 0.2327013, 0.1394811, 0.07206697, -0.1478604, -0.0217588, 1.16737e-06, -0.1378015, -0.1587454, -0.1358493, 0.1394811, 0.0867477, 0.04655997, -0.08567691, -0.01316778, 1.046254e-06, -0.07108649, -0.08189026, -0.07012885, 0.07206697, 0.04655997, 0.02837801, -0.03992732, -0.004572649, 9.503525e-07, 0.1485249, 0.1710188, 0.1463778, -0.1478604, -0.08567691, -0.03992732, 0.09977827, 0.01629235, -1.732561e-07, 0.02089897, 0.02408786, 0.0205581, -0.0217588, -0.01316778, -0.004572649, 0.01629235, 0.004309231, 8.270628e-08, 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p	q	fit	aicbic
4	4	-1.649222, -1.320917, -0.2134461, 0.2361371, 1.124972, 0.401838, -0.6750127, -0.5905637, 0.0002532823, 0.0001876233, 0.1062427, 0.1046864, 0.08437158, -0.01754086, -0.1043899, -0.04485301, -0.02169823, 0.07321954, -3.821729e-07, 0.1046864, 0.1106181, 0.09184523, -0.009910765, -0.1004438, -0.04826142, -0.02667983, 0.06535287, -6.525062e-07, 0.08437158, 0.09184523, 0.07725666, -0.005284638, -0.0800947, -0.04037598, -0.02343931, 0.05027536, -6.192651e-07, -0.01754086, -0.009910765, -0.005284638, 0.01025266, 0.01963982, 0.003380378, -0.001674199, -0.01880398, -2.007239e-07, -0.1043899, -0.1004438, -0.0800947, 0.01963982, 0.104956, 0.04460941, 0.02117627, -0.07415576, 4.89762e-07, -0.04485301, -0.04826142, -0.04037598, 0.003380378, 0.04460941, 0.02486794, 0.01560788, -0.02571369, 7.524306e-07, -0.02169823, -0.02667983, -0.02343931, -0.001674199, 0.02117627, 0.01560788, 0.01186094, -0.008348602, 7.175152e-07, 0.07321954, 0.06535287, 0.05027536, -0.01880398, -0.07415576, -0.02571369, 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p	q	fit	aicbic
5	3	-0.90867, -0.701218, 0.2196466, -0.02729264, -0.02652597, 0.3860108, 0.1567358, -0.7453521, 0.0002537856, 0.0001878682, 0.009632736, 0.0139978, 0.01609874, 0.006713443, 0.003301722, -0.007286985, -0.008098797, -0.007183106, -3.614604e-07, 0.0139978, 0.02386066, 0.02747081, 0.01319135, 0.004781654, -0.01188918, -0.01338512, -0.01181067, -5.412577e-07, 0.01609874, 0.02747081, 0.03442762, 0.01845203, 0.008284242, -0.01448881, -0.01629602, -0.01438474, -5.311125e-07, 0.006713443, 0.01319135, 0.01845203, 0.01194555, 0.00558493, -0.007350567, -0.008231557, -0.007278729, -1.628808e-07, 0.003301722, 0.004781654, 0.008284242, 0.00558493, 0.00401038, -0.003338314, -0.003646167, -0.003256721, -4.039756e-08, -0.007286985, -0.01188918, -0.01448881, -0.007350567, -0.003338314, 0.007429605, 0.008247695, 0.007236739, 4.848676e-07, -0.008098797, -0.01338512, -0.01629602, -0.008231557, -0.003646167, 0.008247695, 0.009337723, 0.008137285, 5.480505e-07, -0.007183106, -0.01181067, -0.01438474, -0.007278729, -0.003256721, 0.007236739, 0.008137285, 0.007291491, 4.828193e-07, -3.614604e-07, -5.412577e-07, -5.311125e-07, -1.628808e-07, -4.039756e-08, 4.848676e-07, 5.480505e-07, 4.828193e-07, 4.817091e-08, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, 1244.79, -2469.58, 5, 3, 0, 0, 12, 0, 0, 0.05740413, 0.02820858, 0.03023175, 0.04152173, -0.01329178, 0.005605241, 0.002156921, -0.01145765, 0.002369436, -0.007664108, 0.02479423, 0.0352677, 0.02490526, 0.01080427, 0.007974006, -0.006407006, -0.01019942, -0.004884555, 0.004764652, -0.008121474, -0.005955117, 0.008439838, -0.007504276, 0.0002072337, 0.01039151, 0.004343479, 0.01052926, 0.001819512, 0.0003600648, -0.00831838, -0.0269995, -0.004403887, -0.01518256, 0.004688953, -0.001835667, -0.02201459, 0.00645579, -0.01279717, -0.006181267, -0.02608335, -0.000156154, 0.003644554, 0.006650799, 0.02498609, -0.002808079, -0.008271421, -0.02343608, 0.004506435, -0.01206488, -0.01128067, -0.004336671, -0.003302603, 0.004111266, 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-0.007581469, -0.00163726, 0.01756388, 0.008089453, -0.01104078, 0.009900078, -0.01910717, 0.01670453, -0.006646258, 0.001020921, -0.004887295, -0.001140701, 0.006436318, -0.01121467, -0.01769162, 0.006921152, -0.002424316, -0.004052867, 0.00674735, 0.00391848, 0.003188012, 0.01516138, -0.0004051743, 0.001663456, 0.004342465, 0.007427707, -0.006130216, -0.01004977	- - 2469.58

```
best_row <- grid %>% filter(!is.na(aic)) %>% slice(1)
best_fit <- best_row$fit[[1]]
best_row
```

```
## # A tibble: 1 x 5
##       p      q fit      aic    bic
##   <int> <int> <list>   <dbl> <dbl>
## 1     4     5 <fc_model> -2477. -2433.
```

We select the model with the lowest AIC among feasible candidates. The estimated parameters are:

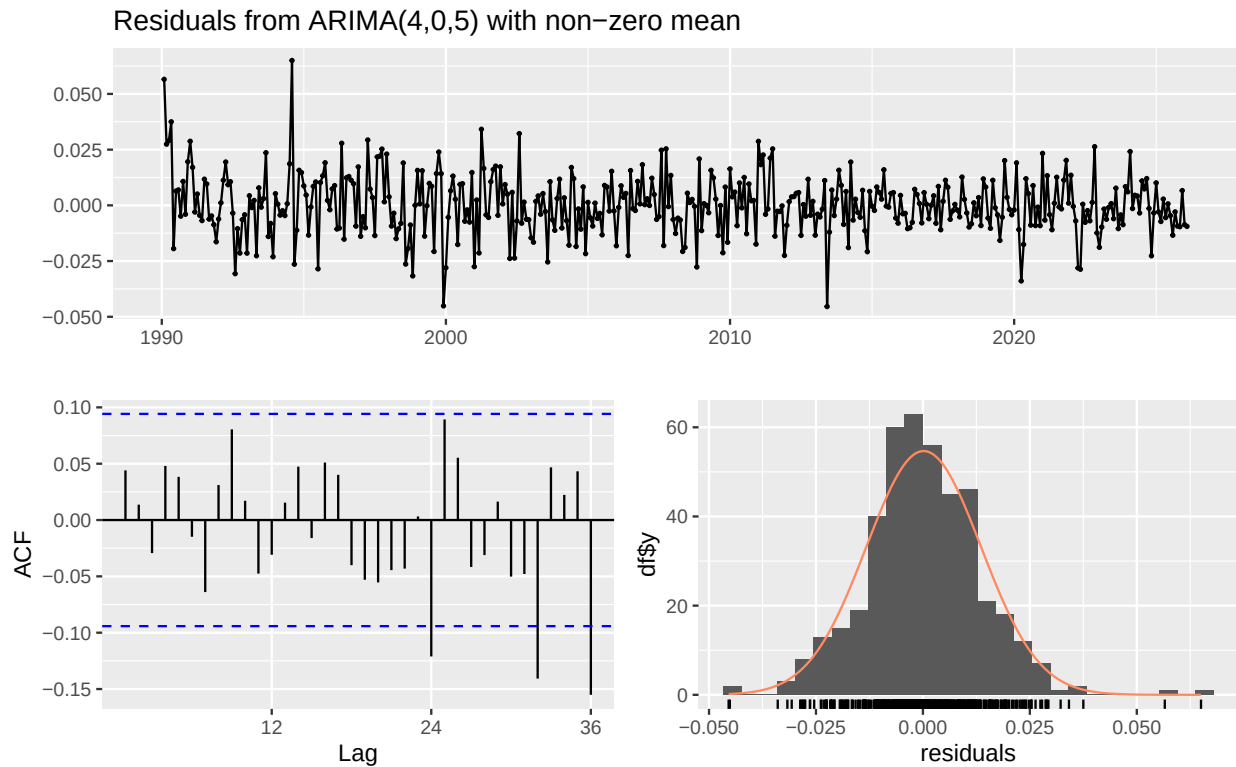
```
summary(best_fit)
```

```
## Series: X
## ARIMA(4,0,5) with non-zero mean
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ma1      ma2      ma3      ma4      ma5
##      0.0541  0.6039  0.0132 -0.9645 -0.625 -0.6444  0.3517  0.9961 -0.6083
## s.e.  0.0161  0.0196  0.0267  0.0157  0.057  0.0457  0.0640  0.0483  0.0708
##      mean
##      3e-04
## s.e.  2e-04
##
## sigma^2 = 0.000184: log likelihood = 1249.69
## AIC=-2477.38   AICc=-2476.76   BIC=-2432.61
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.0001765172 0.01340533 0.01013065 67.12429 239.3093 0.5965699
##              ACF1
## Training set 0.04403442
```

2.1.1 Diagnostic checks

We validate the fitted model via residual autocorrelation, Ljung-Box test, and a Gaussianity check.

```
checkresiduals(best_fit)
```



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(4,0,5) with non-zero mean
## Q* = 24.785, df = 15, p-value = 0.0529
##
## Model df: 9.   Total lags used: 24

res <- residuals(best_fit)
list(
  ljung_box = Box.test(res, lag = 24, type = "Ljung-Box"),
  jarque_bera = tseries::jarque.bera.test(res)
)

## $ljung_box
##
##   Box-Ljung test
##
## data:  res
## X-squared = 24.785, df = 24, p-value = 0.4175
##
##
## $jarque_bera
##
##   Jarque Bera Test
##
## data:  res
## X-squared = 67.782, df = 2, p-value = 1.887e-15
```

If residual diagnostics are satisfactory (no significant autocorrelation, approximate normality), the model is considered valid.

2.2 5. Write the ARIMA(p,d,q) model for the chosen series

Let Z_t be the transformed series (log if used) and $X_t = \nabla^d Z_t$ be the stationary series. If the selected ARMA model is ARMA(p, q) for X_t , then the original series follows an ARIMA(p, d, q) model for Z_t . If $Z_t = \log Y_t$, this corresponds to an ARIMA(p, d, q) for the log series, which implies a multiplicative structure on Y_t .

3 Part III: Prediction

Denote T the length of the series. We assume the residuals are Gaussian.

3.1 6. Confidence region for (X_{T+1}, X_{T+2})

Let $\hat{\mu} = (\hat{X}_{T+1}, \hat{X}_{T+2})^\top$ be the 2-step-ahead forecast mean vector and $\hat{\Sigma}$ the 2x2 forecast error covariance matrix. Under Gaussian residuals, the joint forecast error is bivariate normal, so the level- α confidence region is the ellipse

$$\mathcal{C}_\alpha = \left\{ x \in \mathbb{R}^2 : (x - \hat{\mu})^\top \hat{\Sigma}^{-1} (x - \hat{\mu}) \leq \chi_{2,\alpha}^2 \right\}.$$

For an ARMA model, let $\{\psi_j\}_{j \geq 0}$ be the coefficients of the MA(∞) representation. Then

- $\text{Var}(X_{T+1} - \hat{X}_{T+1}) = \sigma^2$,
- $\text{Var}(X_{T+2} - \hat{X}_{T+2}) = \sigma^2(1 + \psi_1^2)$,
- $\text{Cov}(X_{T+1} - \hat{X}_{T+1}, X_{T+2} - \hat{X}_{T+2}) = \sigma^2\psi_1$,

where σ^2 is the innovation variance.

3.2 7. Hypotheses used to obtain this region

1. The ARMA model for X_t is correctly specified.
2. Innovations are i.i.d. Gaussian with variance σ^2 .
3. Model parameters are treated as known (estimation uncertainty ignored for the region).
4. The process is stationary and invertible (so the MA(∞) representation exists).

3.3 8. Graphical representation for $\alpha = 95\%$

```
alpha <- 0.95

pred <- predict(best_fit, n.ahead = 2)
mu <- as.numeric(pred$pred)

# Extract AR and MA coefficients for psi1
ar <- if (!is.null(best_fit$model$phi)) best_fit$model$phi else numeric(0)
ma <- if (!is.null(best_fit$model$theta)) best_fit$model$theta else numeric(0)
```

```

psi1 <- if (length(ar) == 0 && length(ma) == 0) 0 else ARMAtoMA(ar, ma, lag.max = 1)[1]

sigma2 <- best_fit$sigma2
Sigma <- matrix(
  c(sigma2, sigma2 * psi1, sigma2 * psi1, sigma2 * (1 + psi1^2)),
  nrow = 2
)

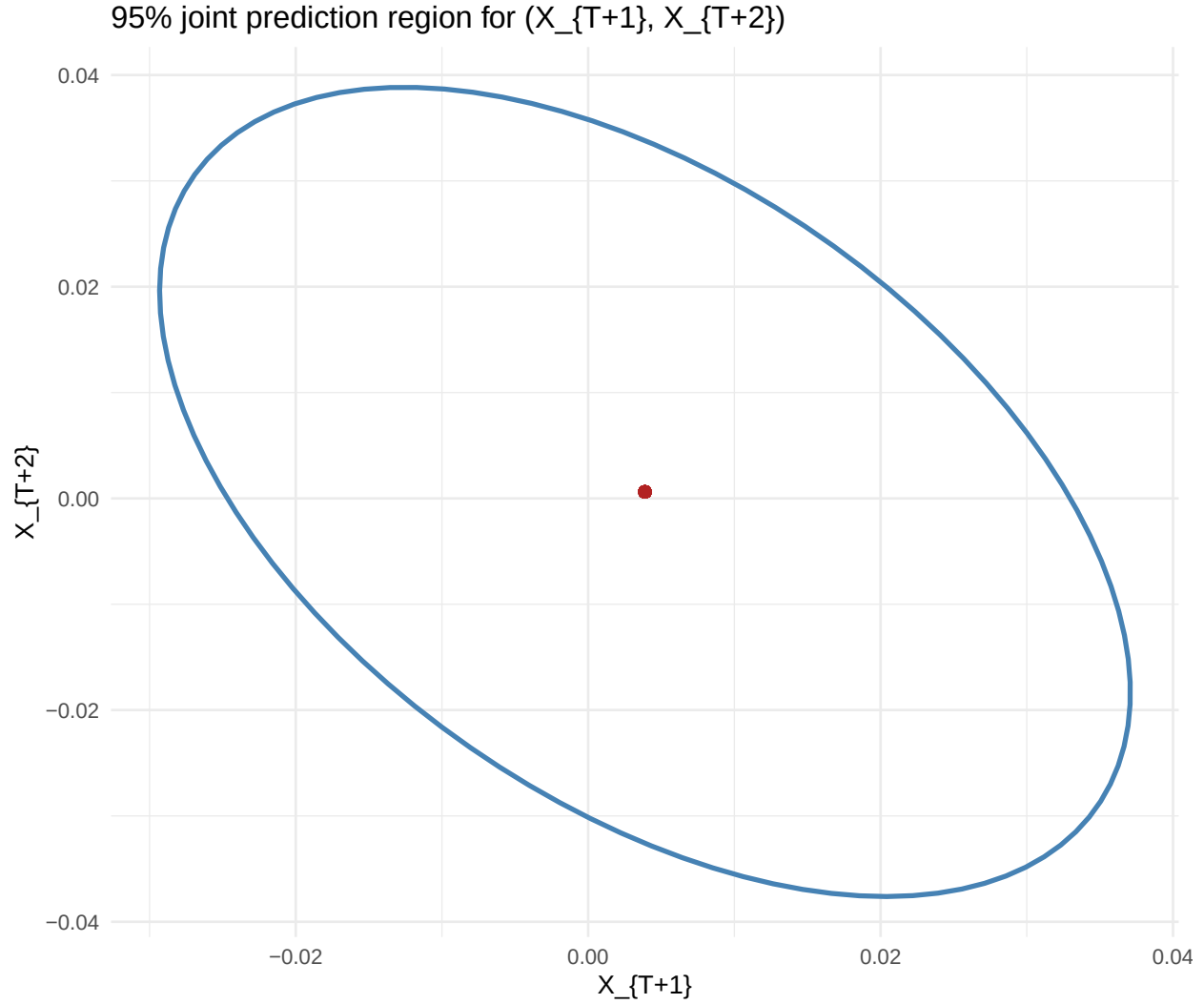
ell <- as.data.frame(ellipse::ellipse(Sigma, centre = mu, level = alpha))
colnames(ell) <- c("X_T1", "X_T2")

corr12 <- Sigma[1, 2] / sqrt(Sigma[1, 1] * Sigma[2, 2])

p <- ggplot(ell, aes(x = X_T1, y = X_T2)) +
  geom_path(color = "steelblue", linewidth = 1) +
  geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 2) +
  labs(
    title = "95% joint prediction region for (X_{T+1}, X_{T+2})",
    x = "X_{T+1}",
    y = "X_{T+2}"
  ) +
  theme_minimal()

```

p



```
corr12
```

```
## [1] -0.4958048
```

Comment: the ellipse orientation reflects the sign of the correlation between forecast errors at horizons 1 and 2. A positive correlation yields an ellipse tilted upward; a negative correlation yields a downward tilt. The size increases with σ^2 and with $|\psi_1|$.

3.4 9. Open question

Let Y_t be a stationary time series observed from $t = 1$ to T , and suppose Y_{T+1} is observed before X_{T+1} . Using Y_{T+1} improves the prediction of X_{T+1} if and only if Y_{T+1} contains incremental predictive information about X_{T+1} beyond the sigma-field generated by $\{X_1, \dots, X_T\}$. Formally, if

$$\mathbb{E}[X_{T+1} \mid X_1, \dots, X_T, Y_{T+1}] \neq \mathbb{E}[X_{T+1} \mid X_1, \dots, X_T],$$

then including Y_{T+1} improves the forecast in mean-square error.

A practical test is to estimate an ARMAX-type model for X_t with Y_t (or a proxy for Y_{T+1} if only available at $T + 1$) and test the significance of the coefficient(s) on Y . This can be done via a t-test on the added regressor(s), or a likelihood-ratio test comparing the ARMA model for X_t against the ARMAX model.